



Pharm*Achieve*

PAIN MANAGEMENT DRUG CHARTS

Qualifying Exam Preparatory Course

LEGEND

AE	Adverse Event	IBD	Inflammatory Bowel Disease
ALT	Alanine Aminotransferase	MAOI	Monoamine Oxidase Inhibitor
ASA	Acetylsalicylic Acid	MI	Myocardial Infarction
BBB	Blood Brain Barrier	MOA	Mechanism of Action
BCRP	Breast Cancer Resistance Protein	N	Nausea
BZD	Benzodiazepine	NSAID	Nonsteroidal Anti-Inflammatory Drug
CABG	Coronary Artery Bypass Graft	OATP	Organic Anion Transporting Polypeptide
CCB	Calcium Channel Blocker	ODT	Oral Disintegrating Tablet
C/I	Contraindication	P-gp	P-Glycoprotein
CNS	Central Nervous System	SJS	Stevens-Johnson Syndrome
CVD	Cardiovascular Disease	SSRI	Selective Serotonin Reuptake Inhibitor
D	Diarrhea	TCA	Tricyclic Antidepressant
D/I	Drug Interaction	TEN	Toxic Epidermal Necrolysis
EPS	Extrapyramidal Symptoms	V	Vomiting
GI	Gastrointestinal		

ACETAMINOPHEN

Drug	MOA	Safety	C/I
Acetaminophen (Tylenol®)	Inhibits prostaglandin synthase in the CNS, analgesic and antipyretic effects	• Liver toxicity • Max daily dose: 4 /day (usually 3 g/day for elderly or chronic users)	• Hypersensitivity to acetaminophen • Acetaminophen-induced liver disease • Excessive alcohol intake

NON-SELECTIVE NSAIDs

DRUG	MOA	SAFETY	C/I
Acetylsalicylic Acid (Aspirin®)	- Non-selectively inhibits COX-1 and COX-2 enzymes - Inhibits COX-1: anti-platelet effects - Inhibits COX-2: analgesic, anti-inflammatory, and antipyretic effects	AEs: - GI upset (N/V/D) - Gastric ulcers, GI bleeds - Renal dysfunction - Tinnitus (ASA) >2400 mg/d of ibuprofen increases risk of serious CV thrombotic events D/I: - Drugs that ↑bleeding risk (anticoagulants, SSRIs, LMWHs, corticosteroids) - Anti-hypertensives - Lithium Metabolism: - Ibuprofen: CYP2C9 (minor: 2C8) - Naproxen: CYP2C9 (minor: 1A2, 2C8)	- ASA only: < 18 years of age with chicken pox, influenza, or flu-like illnesses (risk of Reye's syndrome) - Hypersensitivity to salicylates or NSAIDs 3rd trimester of pregnancy - Active peptic ulcer/bleed, IBD - Severe renal/liver impairment - Severe uncontrolled heart failure - Known hyperkalemia - Peri-operative setting of CABG - Avoid/use cautiously in elderly, cardiac, renal, and GI patients (exception: low-dose ASA in cardiac patients)
Ibuprofen (Advil®)			Same as ASA, but can use in < 18 years of age
Naproxen (Aleve®)			Same as ASA, but can use in > 12 years of age

SEMI-SELECTIVE NSAIDs

DRUG	MOA	SAFETY	C/I
Diclofenac, etodolac indomethacin, meloxicam	Increased affinity for COX-2 but retain COX-1 activity	• Same as ASA • Caution use in patients with CVD history or risk factors • Caution in 1st and 2nd trimesters of pregnancy • Diclofenac (especially at doses ≥ 150 mg/day) has the highest CV risk among semi-/non-selective agents Metabolism • Diclofenac: 2C9 (minor: 2C8, 3A4, 2C18, 2C19, 2B6, 3A4) • Etodolac: CYP2C9 • Indomethacin: 2C9 • Meloxicam: 2C9 (minor: 3A4)	• Same as ASA • Breastfeeding

SELECTIVE NSAIDs

Drug	MOA	Safety	C/I
Celecoxib (Celebrex®)	• Inhibits COX-2: analgesic, anti-inflammatory, & antipyretic effects • Results in lower GI effects • Note, COX-2 selectivity is lost at high doses	• Same as ASA • Doses >200 mg/day increases risk of serious CV events (MI, thrombotic events and stroke) • Caution use in patients with CVD history or risk factors • Caution in 1st and 2nd trimesters of pregnancy • Metabolism: 2C9 (minor: 3A4)	• Same as ASA • Breastfeeding • Allergy to sulfonamides (similar structure)



Reduction in gastroduodenal damage.
Similar efficacy to other NSAIDs.

OPIOIDS

Drug	MOA	Safety	D/I
OPIOIDS (excluding METHADONE) E.g. Morphine Fentanyl Tramadol Oxycodone Hydromorphone Tapentadol	Binding of opioid receptors (mu, delta, kappa) in CNS and intestines. TRAMADOL: Weak mu agonist and 5-HT/NE reuptake inhibitor; M1 metabolite is a strong mu agonist TAPENTADOL: Mu agonist and NE reuptake inhibitor	AEs: • N/V, Constipation • Dizziness • Sedation • Respiratory and CNS Depression • Pruritis • Dry mouth • Cough (fentanyl) • Hyperalgesia • Decreased hormones • Seizures (tramadol)	• CNS depressants (alcohol, BZDs, gabapentin) • QT prolonging drugs (buprenorphine and methadone) • Serotonergic drugs (risk of serotonin syndrome) • CYP2D6 inhibitors, CYP3A4 inducers and inhibitors • CI: MAO inhibitors within 14 days of opioids • CI: Carbamazepine with tramadol (seizure risk)

OPIOID COMBINATIONS

Drug	Indication	Safety	C/I
Codeine (15 mg) with caffeine and acetaminophen (Lenoltec #2®)	Short-term relief of mild to moderate pain	• Drowsiness • Constipation • GI upset	• < 12 years of age • < 18 years of age if tonsillectomy/adenoidectomy • Acute and severe asthma/COPD • Respiratory depression • Any GI obstruction or disease that impacts bowel transit (e.g. ileus) • CYP2D6 polymorphism (codeine) • Pregnant and breast-feeding women • Severe hepatic/renal impairment • MAOI use within last 14 days • Any hypersensitivity to components
Oxycodone (5 mg) with 325 mg acetaminophen (Oxycocet®)	Moderate to moderately severe pain	• Same as above • High abuse potential	
Tramadol (37.5 mg) with 325 mg acetaminophen	Moderate to moderately severe pain	• Same as above • Seizure risks • Caution in elderly (>75y)	

GABA ANALOGS

Drug	MOA	Safety	D/I
GABA Analogs (pregabalin, gabapentin)	↓ neuronal excitability and modulation of nerve impulses	AEs: • Sedation • Weight gain • Dizziness • Peripheral edema • Dry mouth • Potential euphoria and abuse *Renally cleared *Do not abruptly discontinue, taper slowly over 1 week	• Thiazolidinedione anti-diabetic agents • CNS depressants, alcohol

TRICYCLIC ANTIDEPRESSANTS

Drugs	MOA	Safety	D/I
Tricyclic Antidepressants (TCAs): • Amitriptyline • Nortriptyline • Clomipramine • Desipramine • Imipramine • Doxepin	Acts primarily as serotonin-NE reuptake inhibitors	• Anticholinergic effects • Sedation • QT prolongation • Ventricular arrhythmias • Orthostatic hypotension • Weight gain • Sexual dysfunction • Secondary amines have less side effects • Highly toxic in overdose due to cardiac and neuro toxicity	• MAOIs (fatal) • Anticholinergic drugs • Serotonergic drugs (SSRI/SNRIs, tramadol, trazodone, triptans, lithium, linezolid) • CNS depressants • QT-prolonging drugs • Dopamine antagonists (metoclopramide, antipsychotics) • Tamoxifen C/I: recent MI, hypersensitivity Caution: glaucoma, BPH, hx of seizure/CVD/thyroid disorders *CYP1A2, CYP2C19, CYP2D6 substrates



nortriptyline and **desipramine** are secondary amines

TRIPTANS

Drug	MOA	Safety	D/I
- Regular tab - Disintegrating tab - Nasal spray - SC injection	5-HT _{1/2} receptor agonists - Vasoconstriction ↓ excitability of neurons in trigeminal system and decreases inflammation	- Heaviness / tightness / pain in any part of body (e.g. chest, throat, neck, upper limbs) - Elevated BP / palpitations - Nausea / fatigue / malaise - Drowsiness / dizziness - Hypertonia / hypesthesia - Medication overuse HA - Coronary vasospasm - Serious cardiac events (rare) *Risk of serotonin syndrome *Do not use triptans within 24 hours of using another 5-HT₁ agonist (triptan) or ergot-type drug (DHE)	- Ergot-containing drugs - SSRIs / SNRIs - MAOIs (esp. almotriptan, rizatriptan, sumatriptan, zolmitriptan) - CI within 72h of potent CYP3A4 inhibitors (eletriptan) (e.g. ketoconazole, itraconazole, clarithromycin) C/I: History or risk of cardio-/cerebro-/peripheral vascular disease, uncontrolled/severe HTN, severe hepatic impairment, hemiplegic ophthalmoplegic or basilar migraine, hypersensitivity, pregnancy



Triptans that cross BBB: Almotriptan, Naratriptan, Eletriptan, Zolmitriptan

ERGOTAMINE DERIVATIVES

Drug	MOA	Safety	C/I and D/I
Dihydroergotamine (DHE 45) - Injection (IM / IV / SC) - Intranasal spray	5-HT ₁ agonist, acts centrally on dopamine and adrenergic receptors	• N/V • Vasoconstriction (cold, tingling, painful extremities) • Paresthesia / tingling • Chest pain • Muscle cramps • Hypotension (rare) • Medication overuse headache • Rhinitis and taste disturbance (spray)	• Pregnancy • Cardiac conditions • Hypertension • Peripheral Artery Disease • Peptic Ulcer Disease • Sepsis • Liver / kidney disease • Concurrent use with potent CYP3A4 inhibitors D/I: within 24 hours of triptans, MAO, SSRI, SNRI

SEROTONIN ANTAGONISTS

Drug	MOA	Safety	D/I
Pizotifen	5-HT antagonist with antihistamine properties	AEs: • Drowsiness • Weight gain • Anticholinergic effects • Diarrhea • Muscle pain/cramps	• Anticholinergic agents • CNS depressants • MAOIs (CI) • C/I: gastric outlet obstruction

BETA BLOCKERS

Drug	MOA	Safety	D/I
Cardio-selective: bisoprolol, atenolol, metoprolol, nebivolol Non-selective: carvedilol, labetalol, nadolol, propranolol, sotalol	Binds to and blocks β_1 & β_2 receptors	AEs: • Nausea/diarrhea • Hypotension, bradycardia, dizziness, fatigue, headaches • Insomnia / vivid dreams/nightmares • Hyperglycemia, hypoglycemia (may mask symptoms – monitor for sweating) • Bronchospasm, dyspnea • Weight gain • Depression* Discontinue slowly, taper over 1 – 2 weeks	• Anti-hypertensives • Antiarrhythmics • Antihyperglycemics • NSAIDs • CYP1A2/2D6 inhibitors • Beta2-agonist bronchodilators (e.g. formoterol, salbutamol) • Theophylline

NON-DIHYDROPYRIDINE (DHP) CCBS

Drug	MOA	Safety	DI
Verapamil, diltiazem	- Block voltage-gated Ca channels - Increased effect on AV node conduction compared to DHPs	AEs: - Constipation - Anorexia - Nausea ↓BP - Bradycardia; heart block - Gingival hyperplasia	- Strong CYP3A4 inhibitors - Beta-blockers or other cardiac depressant drugs - Carbamazepine, phenytoin - Ivabradine (AVOID) - Dabigatran (Verapamil) - Statins - Rifampin - Digoxin C/I: history/risk of severe hypotension or cardiogenic shock, post-MI with LVEF<40%, AV block, atrial flutter/fibrillation, severe bradycardia

CCB – FLUNARIZINE

DRUG	MOA	SAFETY	D/I
Flunarizine	Selective calcium entry blocker and histamine blocking activity	AEs: • EPS • Galactorrhea • Depression • Drowsiness • Weight gain • Appetite changes	• Sedative drugs (e.g. alcohol, opioids, benzodiazepines) C/I: history of depression, Parkinson's disease, extrapyramidal disorders, hypotension, arrhythmia and heart failure

CALCITONIN GENE-RELATED PEPTIDE (CGRP) INHIBITORS

DRUG	MOA	SAFETY	D/I and C/I
Preventative Monoclonal antibodies: - Erenumab (Aimovig® - SC) - Fremanezumab (Ajovy® - SC) - Galcanezumab (Emgality® - SC) - Eptinezumab (Vyepti® - IV) CGRP inhibitor: - Atogepant (Qulipta® -PO)	Erenumab: antagonizes CGRP receptor, modifies pain receptor signaling and causes vasodilation Fremanezumab: antagonizes CGRP Galcanezumab: antagonizes CGRP Eptinezumab: antagonizes CGRP Atogepant: CGRP receptor antagonist	AEs: - Injection site reactions - Hypersensitivity (C/I) - Severe constipation with complications (erenumab) - Increased blood pressure (erenumab) - Sore throat (eptinezumab) - Nasopharyngitis (eptinezumab, fremanezumab) - Nausea, constipation, decreased appetite, fatigue, somnolence, increased ALT (atogepant) *Dose adj. in mild-mod liver impairment and renal impairment (atogepant)	- Monoclonal antibodies: Not metabolized by CYP enzymes - Atogepant: CYP3A4 (major), OATP Caution due to lack of data: - Pregnancy/breastfeeding - Autoimmune conditions - Liver and renal impairment - CVD conditions - Uncontrolled hypertension (erenumab)

CALCITONIN GENE-RELATED PEPTIDE (CGRP) INHIBITORS

DRUG	MOA	SAFETY	D/I and C/I
Acute treatment • Ubrogapant (Ubrelyv® - PO) • Rimegepant (Nurtec® - ODT)	Antagonizes CGRP receptor	AEs: • Hypersensitivity (C/I) • Nausea (ubrogapant, rimegepant) • Dry mouth (ubrogapant) • Dizziness (rimegepant) • Somnolence (ubrogapant) *Dose adj. in severe renal/liver impairment (ubrogapant)	• CYP3A4 (Rimegepant, ubrogapant) • BCRP/P-gp (ubrogapant, rimegepant) C/I: • Strong CYP3A4 inhibitors (ubrogapant) Caution due to lack of data: • Pregnancy/breastfeeding • Autoimmune conditions • CVD conditions Avoid: • Severe liver impairment (rimegepant) • Dialysis (rimegepant, ubrogapant) • End stage renal disease (rimegepant)

ANTICONVULSANTS – DIVALPROEX / VALPROIC ACID

MOA	Safety	DI
Increased levels of GABA in the brain	AEs: • N/V/D/abdominal cramps/indigestion • Acute pancreatitis • Hepatotoxicity (rare) • Edema (rare) • Gingival hyperplasia • Thrombocytopenia • Sedation • Alopecia • Polycystic ovaries • Teratogenic	↑ Valproate levels: • ASA • Cimetidine • Erythromycin ↓ Valproate levels: • Phenytoin • Rifampin • Carbapenems • Carbamazepine • Protease inhibitors *SJS/TEN with lamotrigine *No effects on oral contraceptives

ANTICONVULSANTS - TOPIRAMATE

MOA	Safety	D/I
Reduces action potential frequency, enhances GABA activity, antagonizes activation of glutamate	AEs: • Upper respiratory tract infection • Headache, fatigue • Weight decrease, anorexia • Somnolence, dizziness • Nausea • Paresthesia • Depression • Fever and trouble concentrating (in pediatrics) • Rare: acute myopia and secondary angle closure glaucoma *Discontinue gradually due to risk of seizures	• Anticonvulsants • CNS depressants • Combined oral contraceptives (decreased estrogen levels → contraceptive failure) • Lithium, risperidone • Amitriptyline C/I: Pregnancy & women of child-bearing age without effective contraception (if only for migraine prophylaxis)

SELECTIVE SEROTONIN REUPTAKE INHIBITORS

Drugs	MOA	Safety	D/I
Escitalopram, citalopram, fluoxetine, fluvoxamine, paroxetine, sertraline	Selectively inhibits serotonin reuptake at presynaptic neuronal membrane	• Nausea, headache, fatigue, yawning, orthostatic hypotension, insomnia, GI effects, nausea, sexual dysfunction, weight gain, anticholinergic effects, • Urinary incontinence, hyponatremia (elderly) • Agitation leading to self-harm/harm to others (reported cases) • Withdrawal syndrome (dizziness, nausea, fatigue, headache and insomnia) at discontinuation or dose reduction	• Serotonergic agents (e.g. MAOI, St. John's wort) • NSAIDs and antiplatelets: ↑ risk of bleeding • SSRIs inhibits some CYP450 enzymes • QT-prolonging drugs • Tamoxifen

SEROTONIN NOREPINEPHRINE REUPTAKE INHIBITORS: DULOXETINE, VENLAFAXINE

MOA	SAFETY	D/I and C/I
Blocks reuptake of 5-HT and NE	• Nausea, headache, dry mouth, insomnia, sweating, sexual dysfunction, fatigue, drowsiness, decreased appetite, constipation • Hepatotoxicity, hyponatremia • Increased BP and HTN (especially venlafaxine) • Agitation leading to self-harm/harm to others (reported cases) • Withdrawal syndrome (dizziness, nausea, fatigue, headache and insomnia) at discontinuation or dose reduction	• Serotonergic agents (e.g. MAOI, triptans) • Anticholinergic agents • Potent CYP1A2 inhibitors (e.g. ciprofloxacin, fluvoxamine) with duloxetine • NSAIDs and antiplatelets: ↑ risk of bleeding Duloxetine C/I: Hepatic impairment, severe renal impairment, end stage renal disease concurrent use with MAOIs • Avoid use in uncontrolled narrow-angle glaucoma Venlafaxine C/I: concurrent use with MAOIs

MUSCLE RELAXANTS: CYCLOBENZAPRINE

MOA	Safety	D/I
Potentiates NE effects, central anticholinergic effects* *Structurally similar to TCAs	AEs: • Drowsiness/fatigue • Dizziness • Anticholinergic effects (e.g. dry mouth) <u>Caution:</u> urinary retention, angle-closure glaucoma, increased intraocular pressure, mild liver impairment	• Acetylcholinesterase inhibitors (e.g. donepezil) • Anticholinergic agents • CNS depressants • CYP1A2 inhibitors • Fluoxetine (Avoid) • MAOIs (CI) • Tramadol (Avoid) • Serotonergic agents C/I: Recent myocardial infarction, hyperthyroidism, history of heart failure, arrhythmias or heart block

MUSCLE RELAXANTS: BACLOFEN

MOA	SAFETY	D/I
Exact MOA: unknown; decreases excitatory neurotransmitters; agonist at GABA receptors	AEs: • Somnolence, dizziness • Muscle weakness • Lowered seizure threshold • Respiratory/CV depression • Abrupt discontinuation can cause confusion, seizures, hallucinations • Visual impairment Caution: renal impairment, severe liver impairment, epilepsy, urinary retention, psychiatric disorders	• CNS depressants (e.g. benzodiazepines, TCAs, opioids) • Levodopa • Anesthetics • Lithium • Anti-hypertensives

REFERENCES

1. Acetaminophen. In: RxTX. Ottawa, ON: Canadian Pharmacists Association. <https://myrxtx.ca/search>
2. Nonsteroidal Anti-inflammatory Drugs (NSAIDs). In: RxTX. Ottawa, ON: Canadian Pharmacists Association. <https://myrxtx.ca/search>
3. Opioids. In: RxTX. Ottawa, ON: Canadian Pharmacists Association. <https://myrxtx.ca/search>
4. Neuropathic Pain. In: RxTX. Ottawa, ON: Canadian Pharmacists Association. <https://myrxtx.ca/search>
5. Tricyclic Antidepressants. In: RxTX. Ottawa, ON: Canadian Pharmacists Association. <https://myrxtx.ca/search>
6. Headache in Adults. In: RxTX. Ottawa, ON: Canadian Pharmacists Association. <https://myrxtx.ca/search>
7. Pizotifen. In: RxTX. Ottawa, ON: Canadian Pharmacists Association. <https://myrxtx.ca/search>
8. Beta-adrenergic Blocking Agents. In: RxTX. Ottawa, ON: Canadian Pharmacists Association. <https://myrxtx.ca/search>
9. Calcium Channel Blockers. In: RxTX. Ottawa, ON: Canadian Pharmacists Association. <https://myrxtx.ca/search>
10. Selective Serotonin Reuptake Inhibitors (SSRIs). In: RxTX. Ottawa, ON: Canadian Pharmacists Association. <https://myrxtx.ca/search>
11. Cymbalta. In: RxTX. Ottawa, ON: Canadian Pharmacists Association. <https://myrxtx.ca/search>
12. Cyclobenzaprine. In: RxTX. Ottawa, ON: Canadian Pharmacists Association. <https://myrxtx.ca/search>
13. Purdy RA. Headache in Adults. E-Therapeutics- Therapeutic choices. Canadian Pharmacists Association.
14. Worthington I. Headache. E-Therapeutics- CTMA. Canadian Pharmacists Association.
15. AbbVie Corporation. QULIPTA (atogepant tablets) [Product monograph]. In QULIPTA (atogepant tablets) (1st ed., pp. 1–30). https://pdf.hres.ca/dpd_pm/00075448.PDF
16. Pfizer Canada ULC. NURTEC ODT (Rimegepant orally disintegrating tablets). In PRODUCT MONOGRAPH [Report]. https://pdf.hres.ca/dpd_pm/00073578.PDF
17. Aimovig. In: RxTX. Ottawa, ON: Canadian Pharmacists Association. <https://myrxtx.ca/search>
18. Emgality In: RxTX. Ottawa, ON: Canadian Pharmacists Association. <https://myrxtx.ca/search>
19. Ajovy In: RxTX. Ottawa, ON: Canadian Pharmacists Association. <https://myrxtx.ca/search>
20. Lundbeck Canada Inc. VYEPTI (Eptinezumab) – Product Monograph. In VYEPTI (Eptinezumab) – Product Monograph (pp. 1–25). https://pdf.hres.ca/dpd_pm/00072914.PDF
21. AbbVie Corporation. UBRELVIY®. In UBRELVIY® (pp. 1–26) [Product monograph]. https://www.abbvie.ca/content/dam/abbvie-dotcom/ca/en/documents/products/UBRELVIY_PM_EN.pdf

CHANGE LOG

May 2025

- Slide 22: Removed thioridazine from D/Is as discontinued

July 2025

- Updated legend
- Content reviewed, no changes made

November 2025

- No content changes made

April 2026

- No content changes made